



Second Week

NLP Internship

Table of Content

Part one : Important Knowledge

- More about LLMs output configurations
- Bert Family
- Introduction to LLaMA family

Part two : plus Knowledge (optional)

- What is an LLM Twin ?
- Why building an LLM Twin Matters ?
- Planning the MVP of the LLM Twin product
- What is an MVP ?
- Defining the LLM Twin MVP ?
- Building ML systems with feature/ training / inference pipelines

Part three : Task

- Task 0
- Task 1

Note: Part two will not be used in the task; it is only provided for your knowledge so you can broaden your understanding.

LLMs Output Configuration

Temperature

Temperature is an important parameter for language models, it influences how random (or “ creative “) the outputs are, it controls the creativity and randomness of the text generated.

When Temperature is low the model make same output each time

$$P_T(w_i) = \frac{e^{\frac{\text{logit}_i}{T}}}{\sum_j e^{\frac{\text{logit}_j}{T}}}$$

Let's say model has calculated these logits (raw scores before applying softmax)

$$\text{loit}(cat) = 2.0, \text{logit}(Dog) = 1.0, \text{logit}(Unicorn) = 0.1$$

Case 1 : Temperature (T = 1.0)

$$P(cat) = \frac{e^{\frac{2.0}{1.0}}}{e^{\frac{2.0}{1.0}} + e^{\frac{1.0}{1.0}} + e^{\frac{0.1}{1.0}}} = 0.659001$$

$$P(Dog) = \frac{e^{\frac{1.0}{1.0}}}{e^{\frac{2.0}{1.0}} + e^{\frac{1.0}{1.0}} + e^{\frac{0.1}{1.0}}} = 0.24243$$

$$P(Unicorn) = \frac{e^{\frac{0.1}{1.0}}}{e^{\frac{2.0}{1.0}} + e^{\frac{1.0}{1.0}} + e^{\frac{0.1}{1.0}}} = 0.098565$$

Case 2 : Temperature (T = 0.3)

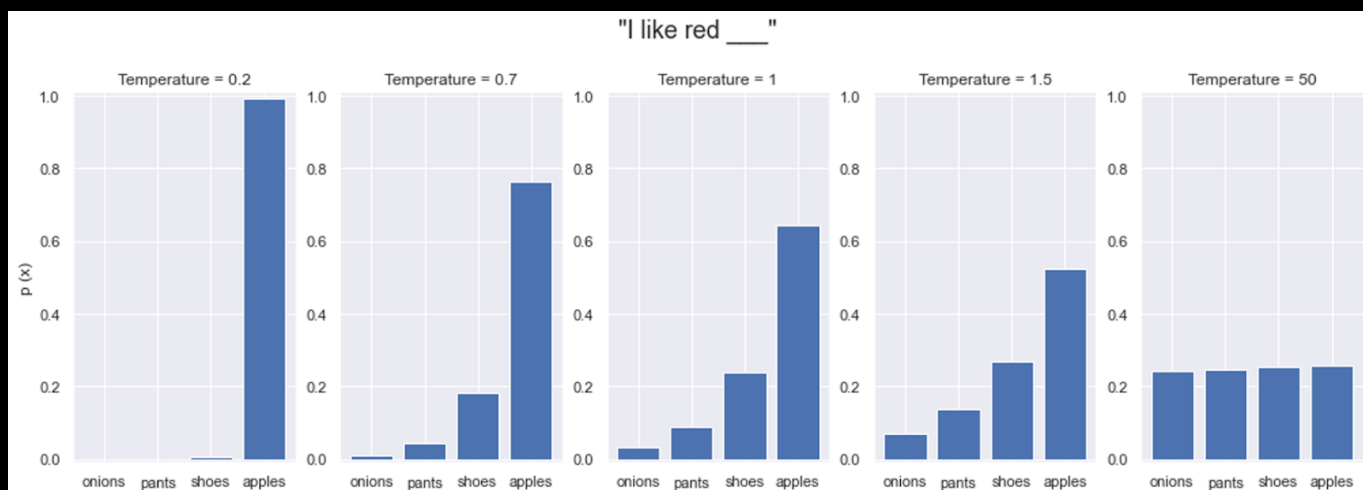
$$P(cat) = \frac{e^{\frac{2.0}{0.3}}}{e^{\frac{2.0}{0.3}} + e^{\frac{1.0}{0.3}} + e^{\frac{0.1}{0.3}}} = 0.96390$$

$$P(Dog) = \frac{e^{\frac{1.0}{0.3}}}{e^{\frac{2.0}{0.3}} + e^{\frac{1.0}{0.3}} + e^{\frac{0.1}{0.3}}} = 0.034386$$

$$P(Unicorn) = \frac{e^{\frac{0.1}{0.3}}}{e^{\frac{2.0}{0.3}} + e^{\frac{1.0}{0.3}} + e^{\frac{0.1}{0.3}}} = 0.001711989$$

Case 3 : Temperature (T = 2.0) [Try it with yourself]

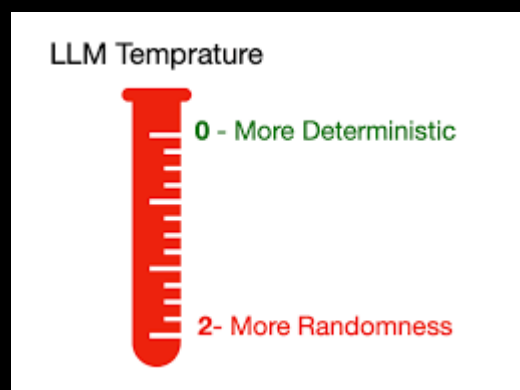
$$\begin{bmatrix} +6.0 \\ +5.0 \\ +4.0 \\ +1.5 \\ +9.9 \\ +2.3 \\ +4.0 \\ +1.9 \end{bmatrix} \rightarrow \begin{bmatrix} e^{x_1/T} / \sum e^{x_n/T} \\ e^{x_2/T} / \sum e^{x_n/T} \\ e^{x_3/T} / \sum e^{x_n/T} \\ e^{x_4/T} / \sum e^{x_n/T} \\ e^{x_5/T} / \sum e^{x_n/T} \\ e^{x_6/T} / \sum e^{x_n/T} \\ e^{x_7/T} / \sum e^{x_n/T} \\ e^{x_8/T} / \sum e^{x_n/T} \end{bmatrix} = \begin{bmatrix} 0.19 \\ 0.02 \\ 0.13 \\ 0.08 \\ 0.42 \\ 0.04 \\ 0.03 \\ 0.09 \end{bmatrix}$$



Read about Temperature from this link :

<https://www.ibm.com/think/topics/llm-temperature>

<https://arxiv.org/pdf/2405.00492>

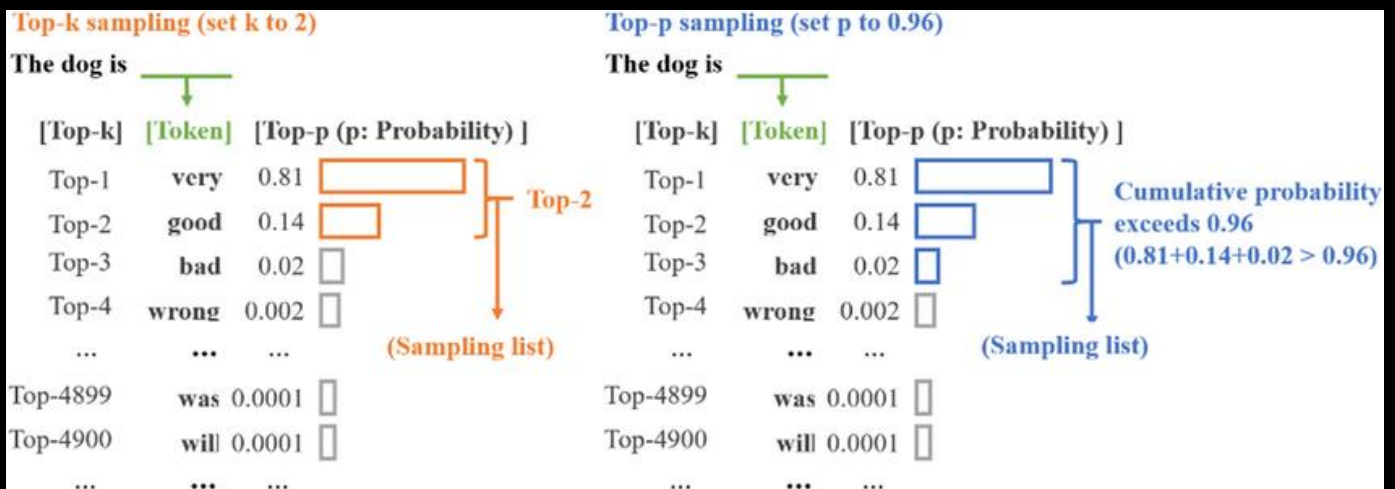


Top – k Sampling

Instead of sampling from all tokens, we keep only the Top k tokens with the highest probabilities and set probability of all others to zero, we then renormalize probabilities so they sum to 1 and sample from this reduced set

Top – P sampling

Sampling selects the top tokens whose cumulative probability does not exceed a certain value (P) , value for P range from 0 to 1 (all tokens in the LLM's vocabulary)



More about top k and top p

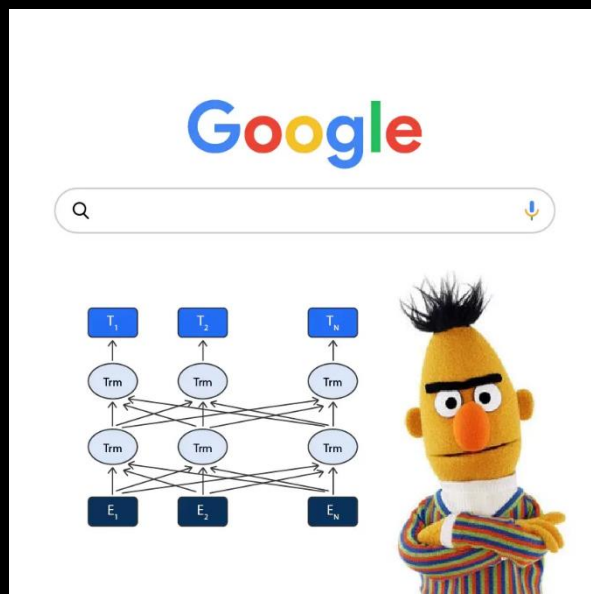
https://youtu.be/_3DWwb96exY?si=sulQiLzGu1DYanbJ

<https://youtu.be/B-W6LjSluCk?si=X4Lt5oczNIIA-88B>

Bert Family

Bert is a big family consists of many models

BERT – Roberta – DistilBert – ALBERT – DeBERTa –
SpanBERT – TinyBERT - XLM-R – ELECTRA



BERT Paper :

<https://arxiv.org/pdf/1810.04805>

Roberta paper :

<https://arxiv.org/pdf/1907.11692>

DistilBert paper :

<https://arxiv.org/pdf/1910.01108>

ALBERT paper :

<https://arxiv.org/pdf/1909.11942>

DeBERTA paper :

<https://arxiv.org/pdf/2006.03654>

SpanBERT paper :

<https://arxiv.org/pdf/1907.10529>

TinyBERT paper :

<https://arxiv.org/pdf/1909.10351>

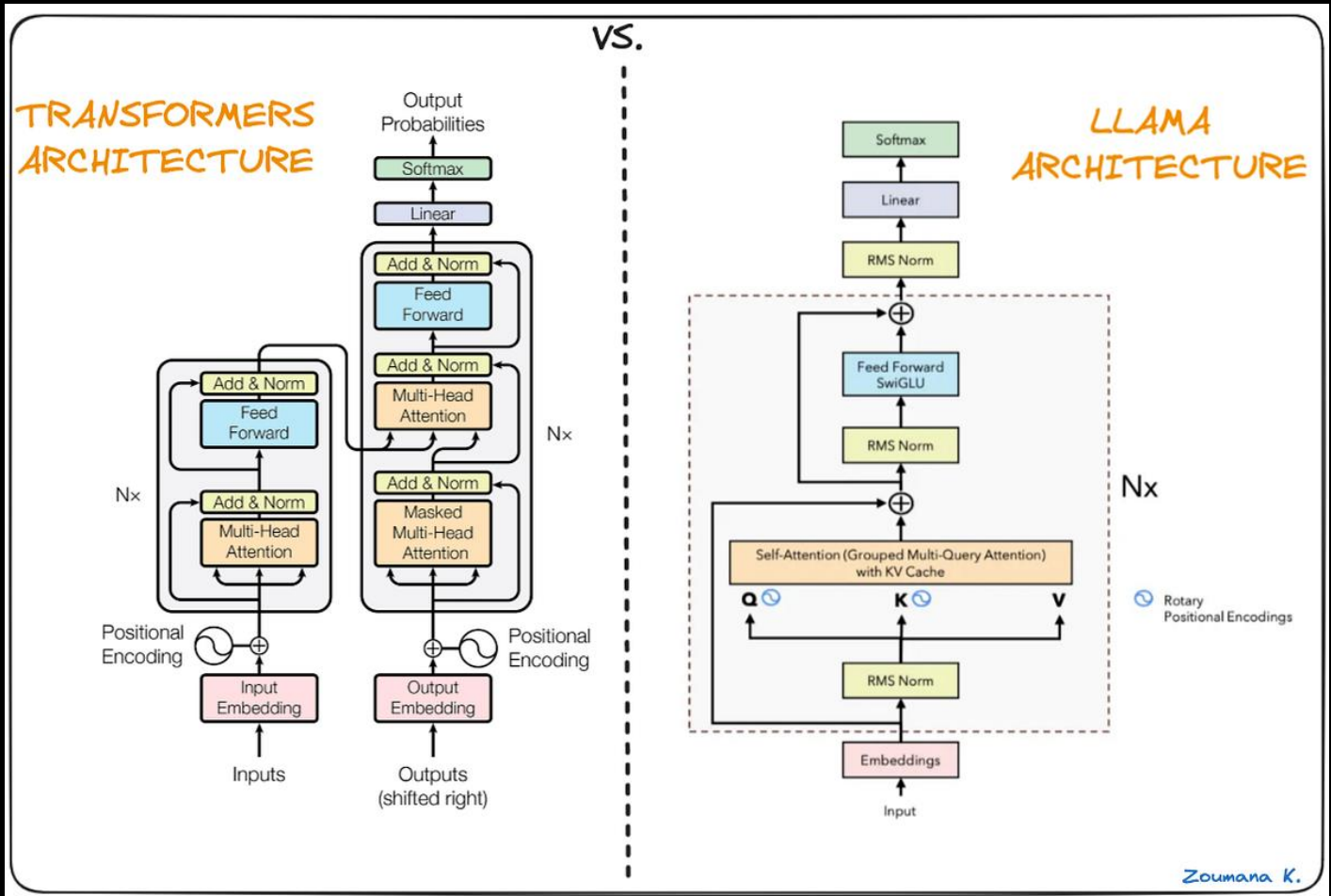
XLM-R paper :

<https://arxiv.org/pdf/1911.02116>

ELECTRA paper :

<https://arxiv.org/pdf/2003.10555>

Introduction to Llama Family



Read this to know more about Llama Architecture

<https://pub.towardsai.net/llama-explained-a70e71e706e9>

[https://en.wikipedia.org/wiki/Llama_\(language_model\)](https://en.wikipedia.org/wiki/Llama_(language_model))

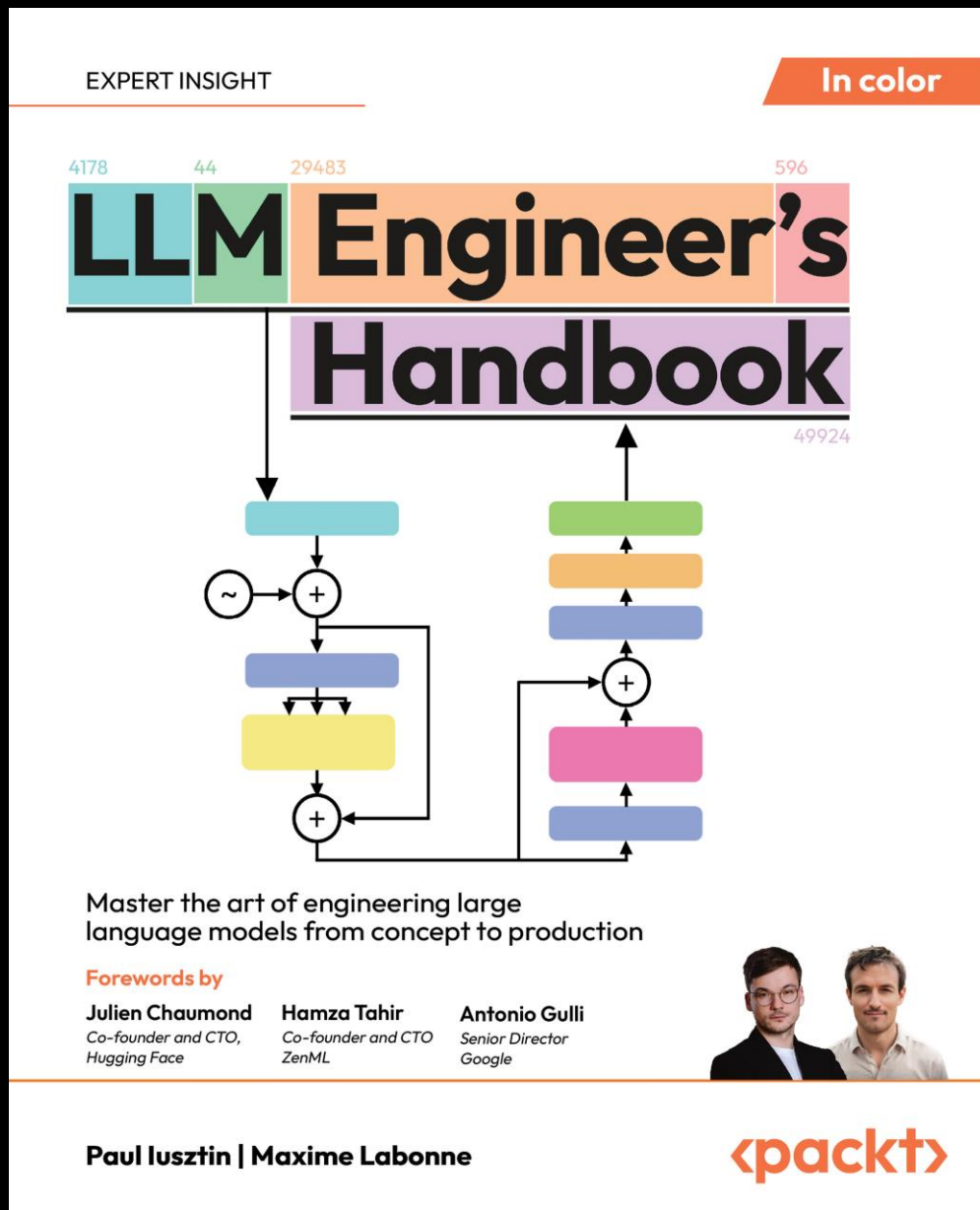
<https://ai.meta.com/blog/meta-llama-3-1/>

watch this video (Important) :

https://youtu.be/Mn_9W1nCFLo?si=d_i9jcry82xqtQ1-

Plus knowledge (LLM Twin) from (LLM Engineering Handbook)

https://drive.google.com/file/d/1K2z-P0D5NXtj8BU-Isi8extcf-Z9ZeSc/view?usp=drive_link



Task

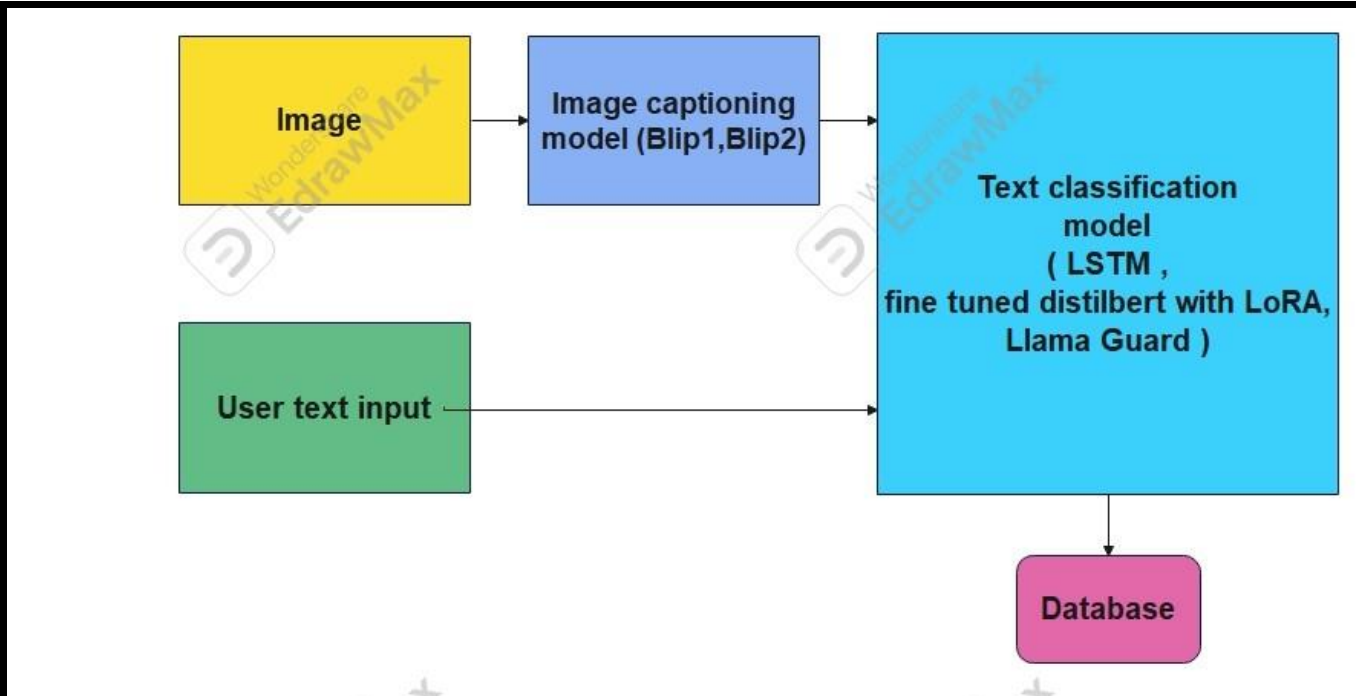
Task 0 : research part

Models like **BERT** or **LLaMA** are very large models. How can we address this issue?

Discuss the **quantization technique** as a solution to reduce model size, and include examples with code.

- Writing research or explanations directly in notebook cells is allowed.
- Using **LaTeX** will earn bonus points.
- Providing **coding examples** will earn bonus points.
- Including **graphs and images** will earn bonus points.

Task 1 : Toxic content classification project



Please adhere strictly to the following requirements for this task

File Format

- Only **.py** files are permitted.
- **.ipynb** (Jupyter Notebook) files are **not** allowed.

Modular Code Structure

- Implement the **image captioning function** in a **separate file** named `imagecaption.py`.
- Import this module into your main script for usage.

Model Selection

- **Image Captioning:** You may use models such as **BLIP-1** or **BLIP-2** from Hugging Face.

Text Classification: Acceptable models include:

- **LSTM**
- **LLaMA Guard**
- **Fine-tuned DistilBERT with LoRA**
- **Fine-tuned ALBERT with LoRA**

Database Management

- Use a **CSV file** as the database node.
- The CSV must **automatically update** whenever the user submits input text or an image caption.

The database should store:

- The **user's text input** or the **generated image caption**.
- The **corresponding classification result**.

Application Framework

- Develop the application using **Streamlit**.

Database Viewing Option

- Include a feature that allows viewing **all stored user inputs and their classifications** from the database at any time.

